Beyond clinical scales: Guiding treatment planning and assistive device selection after spinal cord injury using the SCI-GDI-like score

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Introduction

Walking recovery after incomplete spinal cord injury (iSCI) is commonly monitored using clinical measures which describe walking independence, speed, and lower-limb strength, but not the underlying biomechanics of gait [1]. Instrumented gait analysis can complement clinical assessment by quantifying spatiotemporal and kinematic features of walking. The Gait Deviation Index (GDI) summarizes global kinematic gait deviation with respect to a reference gait pattern [2], and SCI-specific adaptations have been proposed to better characterize gait deviations in this population [3]. This study explored whether a SCI-GDI-like score can provide complementary information for treatment planning during subacute iSCI rehabilitation.

Methods

Six individuals with iSCI were assessed at Institut Guttmann (Badalona, Spain) across four rehabilitation sessions approximately one month apart. Clinical outcomes included WISCI-II, 10-meter walking test (10MWT)-derived gait speed, and lower-limb manual muscle testing (MMT). Wearable inertial sensors (Xsens Awinda, Xsens Technologies B.V., Enschede, The Netherlands) recorded walking trials under feasible assistive (crutches, walker) or orthotic conditions. A local SCI-GDI-like score was computed from segmented pelvis and lower-limb kinematics and referenced to a local healthy-subject dataset, with values closer to 100 indicating greater similarity to healthy gait [2,3]. The score was analysed by subject, session, and walking condition. For comparison with clinical outcomes, one representative walking condition was selected for each subject and session.

Results

Clinical gait speed and the SCI-GDI-like score showed a positive association ($\rho = 0.80$; Fig.1), indicating that higher walking speed was generally accompanied by greater similarity to the healthy-reference gait pattern. However, the dispersion between subjects suggests that clinical walking performance and global kinematic gait quality were not equivalent. Within-session comparisons also showed condition-dependent variability in the SCI-GDI-like score (Fig.2). In subjects B, C, D and E different walking conditions within the same rehabilitation session produced different score ranges, showing that assistive devices and orthotic configurations influenced the observed gait pattern.

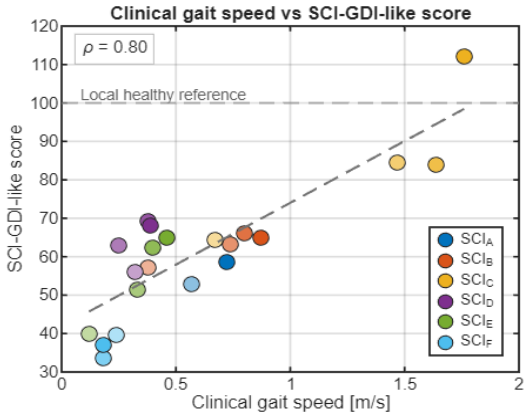


Figure 1: Relationship between clinical gait speed and SCI-GDI-like score using clinically matched walking conditions. Colours identify subjects; lighter-to-darker markers indicate session progression.

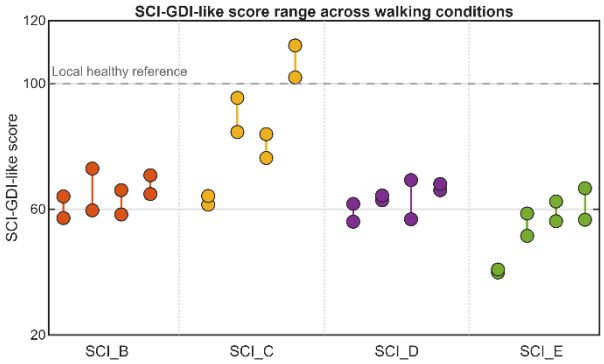


Figure 2: Condition-dependent SCI-GDI-like score variability in four selected subjects. Each vertical segment shows the range between the lowest and highest scores within the same session.

Conclusions

These results suggest that the SCI-GDI-like score complements information from clinical outcomes during subacute iSCI rehabilitation. While clinical scales describe functional capacity, speed, and strength, the SCI-GDI-like score summarizes global kinematic gait quality and captures condition-specific differences across walking configurations.

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References

1. Dittuno and Ditunno, Spinal Cord, 39:654-656, 2001.
2. Schwartz and Rozumalski, Gait Posture, 28:351-357, 2008.
3. Herrera-Valenzuela et al, Front Bioeng Biotechnol, 10:874074, 2022.